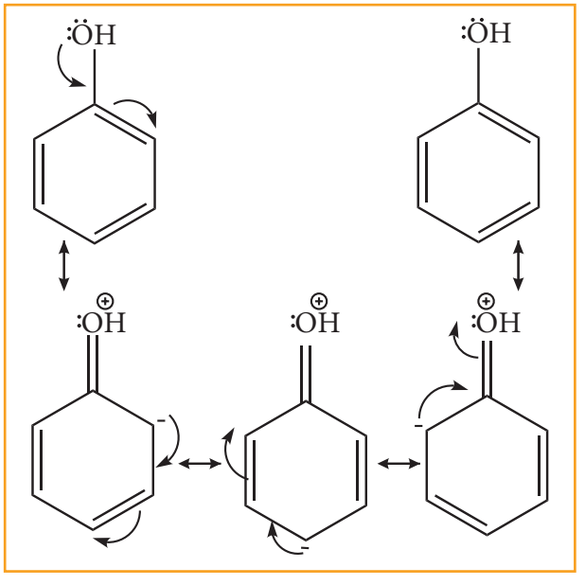
these groups into two categories depending on the way they influence the orientation of attack by the incoming groups. Those which increases electron density at 'ortho' and 'para' position are known as ortho-para directors while those which increase electron density at 'meta' position is known as meta-directors. Some examples of directive influence of functional groups in mono-substituted benzene are explained below.

Ortho and para directing groups

All the activating groups are 'ortho-para' directors. Example –OH, $-NH_2$, -NHR, $-NHCOCH_3$, $-OCH_3$-CH_3 – C_2H_5 etc. Let us consider the directive influences of phenolic (-OH) group. Phenol is the resonance hybrid of following structures.

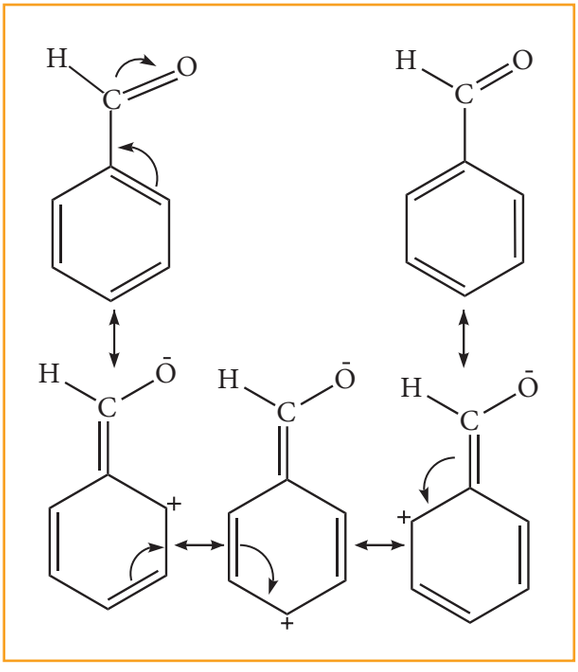


In these resonance structures, the (-) charge residue is present on ortho and para position of ring structure. It is quite evident that the lone pair of electron on the atom which is attached to the ring involves in resonance and makes the ring more electron rich than benzene. The electron density at ortho and parapositions increases as compared to the meta position. Therefore phenolic group activates the benzene ring for electrophilic attack at 'ortho' and 'para positions and hence –OH group is an ortho-para director and activator.

In aryl halides, the strong –I effect of the halogens (electron withdrawing tendency) decreases the electron density of benzene ring, thereby deactivating for electrophilic attack. However the presence of lone pair on halogens involved in the resonance with pi electrons of benzene ring, increases electron density at ortho and para positions. Hence the halogen group is an ortho-para director and deactivator.

META DIRECTING GROUPS

Generally all deactivating groups are meta-directors. For example $-NO_2$, -CN, -CHO, -COR, -COOH, -COOR, $-SO_3H$ etc. Let us consider the directive influence of aldehyde (-CHO) group. Benz aldehyde is the resonance hybrid of following structures.



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